

UNIT 07

CHEMICAL EQUILIBRIUM

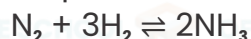


CHEMICAL EQUILIBRIUM:

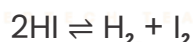
Chemical equilibrium is a state in which the rates of forward and reverse reactions in a chemical system are equal, resulting in a constant concentration of reactants and products.

Example:

1) The reaction between nitrogen gas (N_2) and hydrogen gas (H_2) to form ammonia (NH_3) exhibits chemical equilibrium:



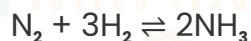
2) The dissociation of hydrogen iodide (HI) into hydrogen (H_2) and iodine (I_2) is another example of chemical equilibrium:

**Reversible reactions:**

Reversible equilibrium refers to a state in which a chemical reaction can proceed in both the forward and reverse directions. It occurs when the rates of the forward and reverse reactions are equal.

Example:

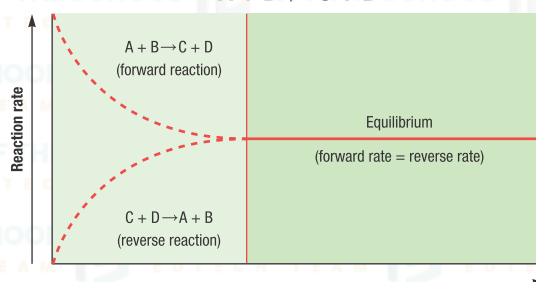
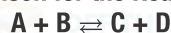
Consider the reaction between nitrogen gas (N_2) and hydrogen gas (H_2) to form ammonia (NH_3):

**Dynamic Equilibrium:**

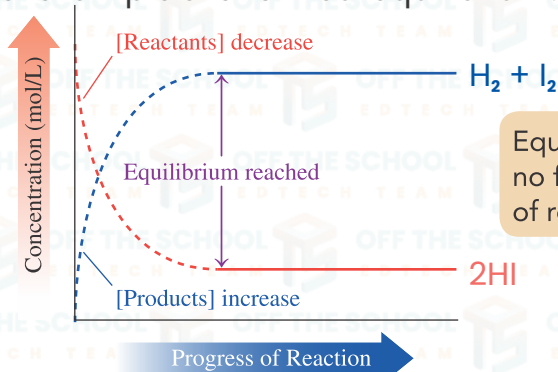
Dynamic equilibrium describes a situation in which the concentrations of reactants and products remain constant over time, while the individual molecular-level processes continue to occur. It is characterized by a balance between forward and reverse reactions.

Equilibrium Reaction

Systems Shown are reaction rates for the hypothetical equilibrium reaction system $A + B \rightleftharpoons C + D$. From the time A and B are mixed together at t_0 , the rate of the forward reaction declines and the rate of the reverse reaction increases until both forward and reverse reaction rates are equal at t_1 , when the equilibrium condition begins.

Rate Comparison for the Reaction System**Example:**

Consider the dissociation of hydrogen iodide (HI) into hydrogen (H_2) and iodine (I_2) is another example of chemical equilibrium:



Equilibrium is reached when there are no further changes in the concentrations of reactants and products.

Macroscopic Characteristics of Forward and Reverse reaction in Reversible Equilibrium:

In a reversible equilibrium, the forward and reverse reactions exhibit the following macroscopic characteristics:

Forward Reaction:

1. The forward reaction proceeds from reactants to products.
2. The concentration of reactants gradually decreases over time.



Reverse Reaction:

1. The reverse reaction proceeds from products to reactants.
2. The concentration of products gradually decreases over time.
3. The concentration of reactants increases as the reaction proceeds.

Macroscopic Characteristics of Forward and Reverse in Dynamic Equilibrium:

In a dynamic equilibrium, the forward and reverse processes exhibit the following macroscopic characteristics:

Forward Process:

1. The forward process involves the conversion of reactants into products.
2. The concentration of reactants decreases as the forward process occurs.
3. The concentration of products increases as the forward process takes place.

Reverse Process:

1. The reverse process involves the conversion of products back into reactants.
2. The concentration of products decreases as the reverse process occurs.
3. The concentration of reactants increases as the reverse process takes place.

H₂, I₂, HI Equilibrium System

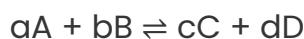
The violet color of the reactant iodine gas, I₂, gives an indication of the progress of the reaction between hydrogen and iodine. Both H₂ (reactant) and HI (product) are colorless gases. The darker the purple color in the flask, the less I₂ gas reacted, so the more reactants remain.

**Law of Mass Action and Its Derivation:****Introduction:**

It was formulated independently by Cato Guldberg and Peter Waage in 1864.

Statement:

According to the law of mass action, the rate of a chemical reaction is directly proportional to the product of the active masses of the reactants each term raised to its stoichiometric coefficients.



Now, from the law of mass action, we know that the rate of forward reaction (R_f) and rate of backward reaction (R_b) will be:

$$R_f = k_f [A]^a [B]^b \text{ --- (i)}$$

$$R_b = k_b [C]^c [D]^d \text{ --- (ii)}$$

Where k_f and k_b are the rate constants for the forward and backward reactions, respectively. After equilibrium is reached, we have

$$R_f = R_b$$

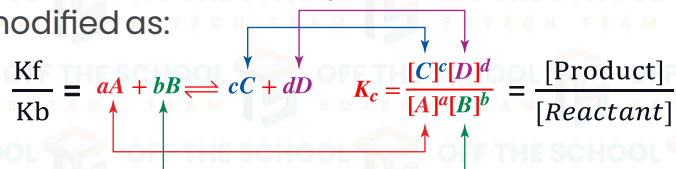
$$k_f [A]^a [B]^b = k_b [C]^c [D]^d$$

$$\frac{k_f}{k_b} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$



Since the K_f and K_b are also constant at equilibrium, the ratio of the two is also a constant and is typically labelled as K_c or the equilibrium constant.

Therefore, equation (i) is modified as:



All this leads to the modern definition of the “law of mass action” that the ratio of the multiplication of molar concentrations of products raised to the power of their stoichiometric coefficients to the multiplication of the molar concentrations of the reactants raised to the power of their stoichiometric coefficients is constant at constant temperature and is called as “equilibrium constant”. This equation is also known as the “law of chemical equilibrium”.

Equilibrium Constant Expression:

Here are some general characteristics of the equilibrium constant expression:

- 1) K_c only works in equilibrium.
- 2) Important characteristics of equilibrium constant expression are as follows K_c only works in equilibrium
- 3) It represents the equilibrium concentration of the reactant and product in mol.dm^{-3} . K_c varies with temperature.
- 4) K_c is independent of reactant and product concentrations.
- 5) K_c is a balanced chemical equation coefficient. In a balanced chemical equation each reactant and product have a concentration equal to its coefficient
- 6) K_c represents equilibrium position. If K_c is larger than 1, the reaction is forward. If K_c is less than 1, the reaction is a reverse reaction.
- 7) Remember that equilibrium constant K_c is a ratio of reactant to product that is utilized to define chemical behavior.

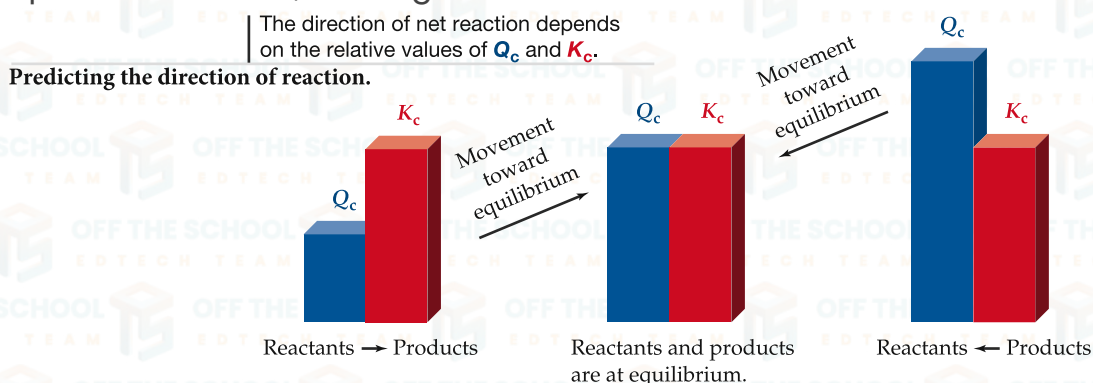
Importance of equilibrium constant:

The value of K_c varies depending on the response. K_c isn't only a calculating constant. It affects both the direction and the extent of a chemical reaction.

1. Direction of a chemical reaction:

Determining the direction of reversible reactions is crucial for process optimization. The reaction quotient, Q_c , helps predict the reaction's state at any given time using actual concentrations. We have three categories:

1. If $Q_c = K_c$, the actual product and reactant concentrations are equal to the equilibrium concentrations, and the system is stable.
2. If $Q_c < K_c$ then there is increase in product concentration for equilibrium. So the forward reaction occurs, forming additional products.
3. If $Q_c > K_c$, there is decrease in product concentration & to achieve equilibria As, the process reverses, forming more reactants.



Movement toward equilibrium changes the value of Q_c until it equals K_c , but the value of K_c remains constant.

2. Extent of chemical reaction:

At a certain temperature, the extent of a reaction measured. The magnitude of an equilibrium constant can predict the scope of a chemical reaction. As magnitude may be very high, very low, or moderate, so can be Extent of chemical reaction.

i. K_c is very small:

Reactions with low K_c never finish. That is, maximum reactant concentration and minimum product concentration. These are called 'reverse or backward responses.



ii. K_c is very large:

Reactions with high values are virtually complete. That is, maximum product concentration and minimum reactant concentration. This type of reaction is known as 'Forward reaction'.



iii. K_c neither very small nor very large:

Reactions which have moderate value of K_c are considered to be at equilibrium. The concentration of reactants and products is almost same. For example:



LE-CHATELIER'S PRINCIPLE AND CHEMICAL EQUILIBRIUM:

Introduction:

The principle is credited to French chemist Henry Louis Le Chatelier and sometimes also to German scientist Karl Ferdinand Braun, who discovered it independently.

Statement:

Le Chatelier's Principle states that if a dynamic equilibrium is disturbed by changing the conditions, such as concentration, temperature, or pressure, the system will adjust itself to counteract that change.

1) Concentration

Le Chatelier's principle explains how chemical equilibria react to changes. Using the $CO + 2H_2 \rightleftharpoons CH_3OH$, reaction as an example, if we increase the amount of **CO** (carbon monoxide), the reaction makes more **CH₃OH** (methanol) to balance this change. Similarly, increasing **H₂** (hydrogen) also increases methanol production.

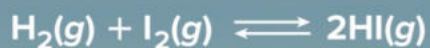
On the contrary, decreasing **CO** or **H₂** shifts the balance towards breaking down methanol into its original reactants.

2) Pressure:

The effect of change in pressure on a system in chemical equilibrium is a key aspect of Le Chatelier's Principle. This principle states that if a system at equilibrium is subjected to a change in one of the conditions, such as pressure, the system adjusts itself to partially counteract the effect of the applied change. Here's how pressure changes can affect a system in equilibrium:



Effect of Addition to an Equilibrium Mixture



$$Q_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(0.825)^2}{(0.112)(0.112)} = 54.3$$

$$Q_c = K_c$$



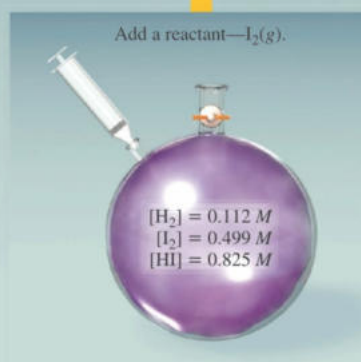
$$Q_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(0.825)^2}{(0.112)(0.112)} = 54.3$$

$$Q_c = K_c$$



$$Q_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(0.825)^2}{(0.112)(0.112)} = 54.3$$

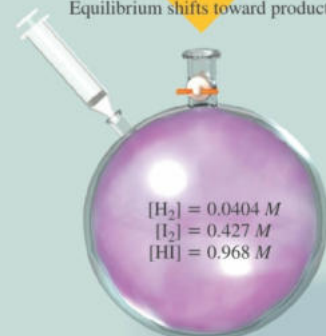
$$Q_c = K_c$$



$$Q_c = \frac{(0.825)^2}{(0.112)(0.499)} = 12$$

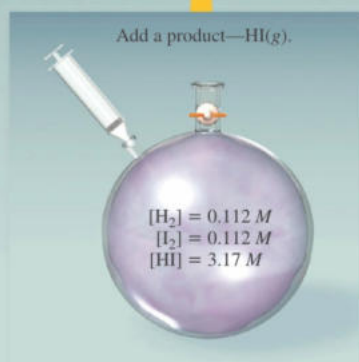
$$Q_c \neq K_c$$

Equilibrium shifts toward product.



$$Q_c = \frac{(0.968)^2}{(0.0404)(0.427)} = 54.3$$

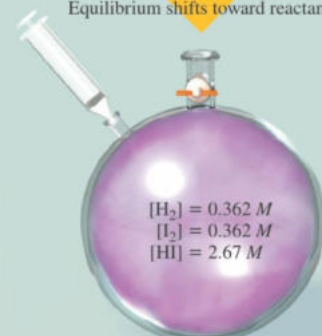
$$Q_c = K_c$$



$$Q_c = \frac{(3.17)^2}{(0.112)(0.112)} = 801$$

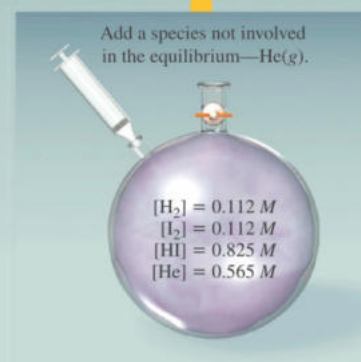
$$Q_c \neq K_c$$

Equilibrium shifts toward reactant.



$$Q_c = \frac{(2.67)^2}{(0.362)(0.362)} = 54.4$$

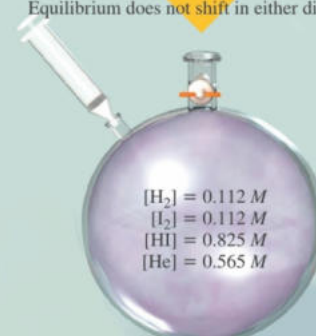
$$Q_c = K_c$$



$$Q_c = \frac{(0.825)^2}{(0.112)(0.112)} = 54.3$$

$$Q_c = K_c$$

Equilibrium does not shift in either direction.



What's the point?

Adding a reactant to an equilibrium mixture shifts the equilibrium toward the product side of the equation. Adding a product shifts the equilibrium toward the reactant side. Adding a species that is neither a reactant nor a product does not cause a shift in the equilibrium.

Effect of Temperature Change



Temperature decrease drives an endothermic equilibrium toward reactants.

Temperature increase drives an endothermic equilibrium toward products.



Temperature increase drives an exothermic equilibrium toward reactants.

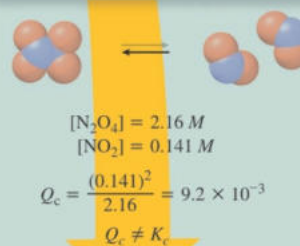
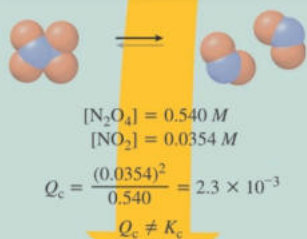
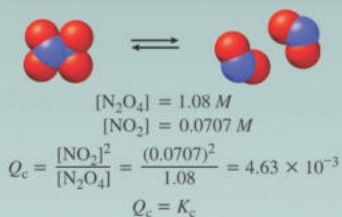
Temperature decrease drives an exothermic equilibrium toward products.



What's the point?

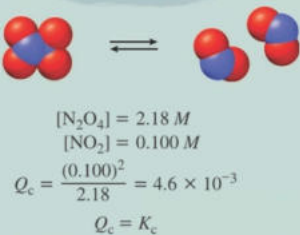
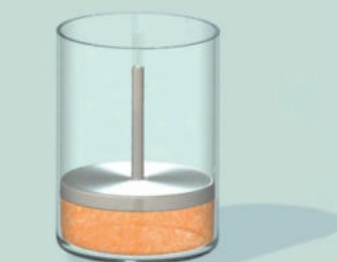
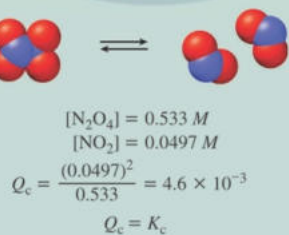
Increasing the temperature of an equilibrium mixture causes a shift toward the product side for an endothermic reaction, and a shift toward the reactant side for an exothermic reaction.

Effect of Volume Change



What's the point?

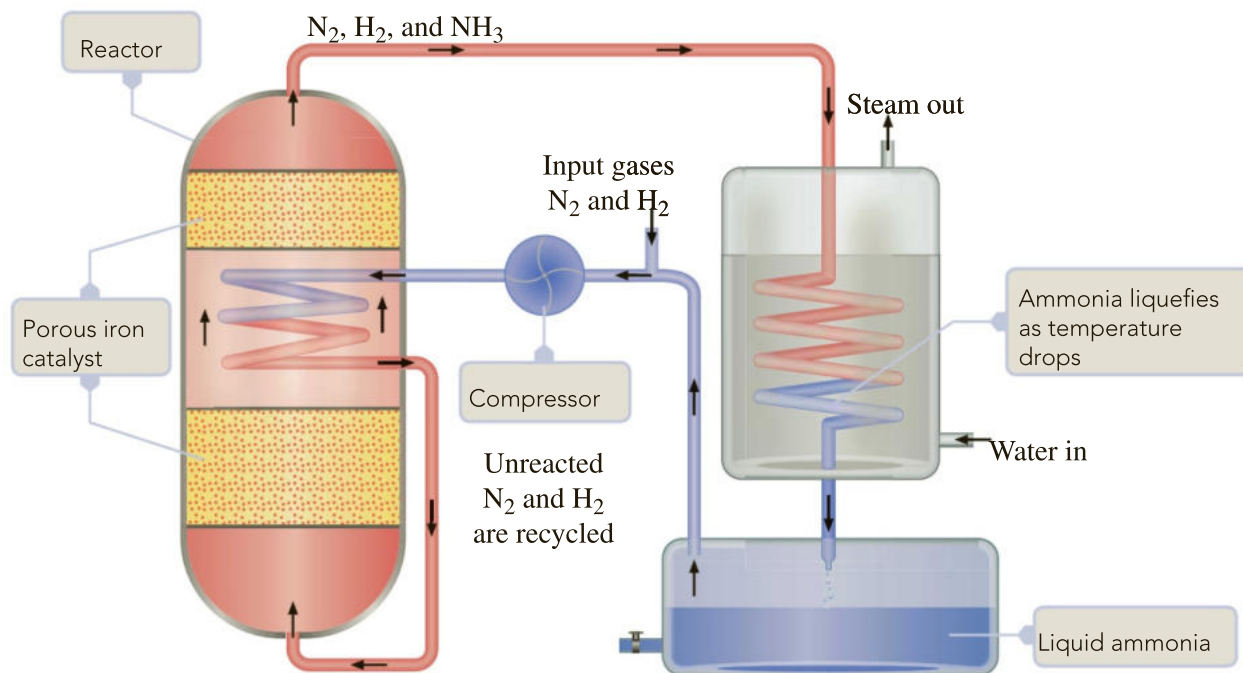
Increasing the volume causes a shift toward the side with the *larger* number of moles of gas.
Decreasing the volume of an equilibrium mixture causes a shift toward the side of the equation with the *smaller* number of moles of gas.



Le Chatelier's Principle and Catalysts

Le Chatelier's principle does not apply to catalysts. Adding a catalyst does not shift the equilibrium of a chemical reaction because it increases the rates of the forward and reverse reactions equally.

Application of Le Chatelier's principle applied to the ammonia formation reaction:



▲ **Industrial production of ammonia** Shown is a schematic diagram of the production of ammonia from nitrogen (obtained from air) and hydrogen (from natural gas). Input gases (N_2 and H_2) are compressed to about 200 atm and enter the reactor (the vessel on the left) where they are heated by the exothermic reaction mixture to about 450°C . The hot reaction gas then passes through the porous iron catalyst to give an equilibrium mixture of N_2 , H_2 , and NH_3 . This mixture exits the reactor and goes to a vessel where it is cooled by water. In the process, steam is produced, which is used to power the compressor and other parts of the ammonia plant. When the reaction gases are sufficiently cooled (at high pressure), the ammonia liquefies. The unreacted N_2 and H_2 are recycled back to the reactor.

Change Type	Effect on Reaction Equilibrium
Concentration Increase	Shifts away from added substance
Concentration Decrease	Shifts towards removed substance
Pressure Increase	Shifts towards fewer gas molecules (NH_3)
Pressure Decrease	Shifts towards more gas molecules (N_2 and H_2)
Temperature Increase	Favors the endothermic reaction (NH_3 formation)
Temperature Decrease	Favors the exothermic reaction (N_2 and H_2 formation)

SOLUBILITY:

The amount of salt dissolved in one liter of a solution at a given temperature to form a saturated solution.

Unit: g/dm³ or g/L.

SPARINGLY SOLUBLE SALTS:

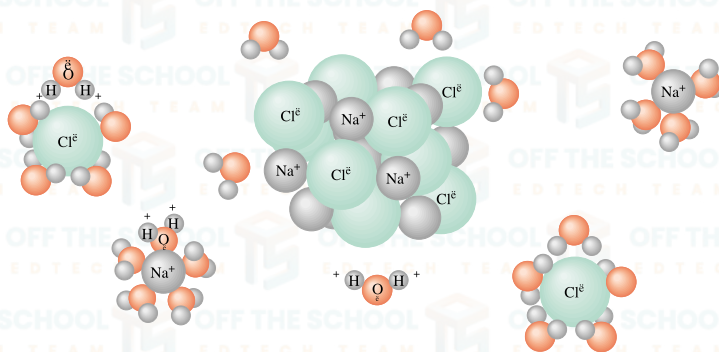
Ionic compounds are slightly soluble in water are called sparingly soluble salt.

Example: Ca₃(PO₄)₂, AgCl, BaSO₄ etc.

SOLUBILITY PRODUCT:

The product of the molar concentration of positive & negative ions of sparingly soluble salts, each raised to the power of its coefficient in the ionized equilibrium equation.

Unit: mol²/dm⁶ or mol³/dm⁹



▲ **Process of dissolving NaCl (salt) in water:** The image shows how water molecules separate the Na⁺ and Cl⁻ ions from the salt crystal and attract them to complete the dissolution process.

Applications of K_{sp}:**1. What is a Precipitation Reaction?**

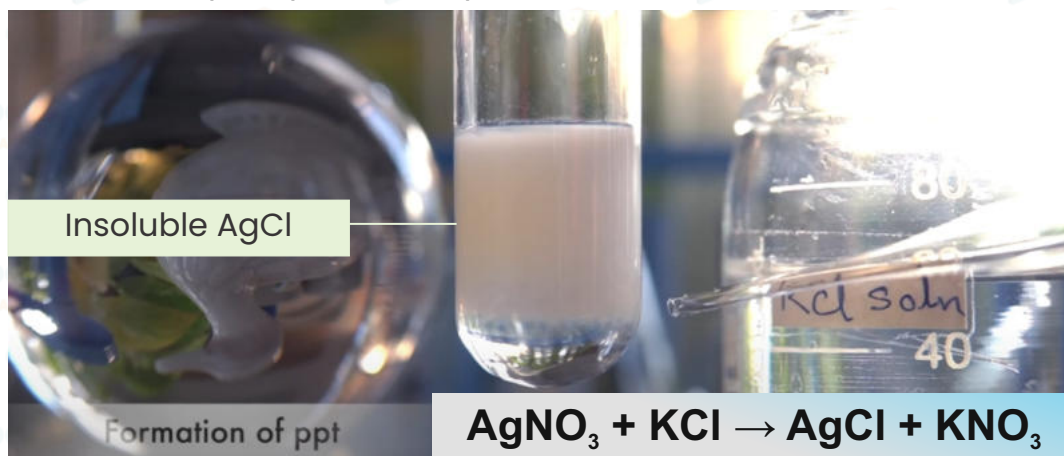
• It's a type of reaction where two solutions combine, and an insoluble solid, called a precipitate, forms and falls out of the solution.

2. What is a Precipitate?

• A substance that doesn't dissolve in water, forming a solid at the bottom of a liquid mixture.

3. Example of a Precipitation Reaction:

• Mixing silver nitrate (AgNO₃) with potassium chloride (KCl) forms silver chloride (AgCl), which is the precipitate, and potassium nitrate (KNO₃) in the water.



4. Ionic Equation:

• The equation only shows the particles that change during the reaction. For the example above, it only shows the silver (Ag^+) and chloride (Cl^-) ions coming together to form silver chloride solid ($\text{AgCl}_{(s)}$).

5. What is Ksp?

• The solubility product constant (K_{sp}) is a number that tells us how much of a compound can be dissolved in water.

6. What does Ksp Indicate?

• A higher K_{sp} means more of the substance can dissolve; a lower K_{sp} means less can dissolve.

7. Ionic Product (Q_{sp}) vs. Solubility Product (K_{sp}):

• The ionic product is like a "test score" of how much ion is actually in the solution at any moment. We compare it to the "passing score" (K_{sp}) to see what happens with our precipitate.

8. How to Predict What Happens in the Solution?

- If the ionic product Q_{sp} is less than K_{sp} , no precipitate forms.
- If the ionic product Q_{sp} is equal to K_{sp} , the solution is just right, and it's at equilibrium.
- If the ionic product Q_{sp} is more than K_{sp} , too much solid has formed, and it's a supersaturated solution.

9. Interesting Fact:

• Some compounds are not very soluble but are very important. For example, barium sulphate is used in medical imaging because it's opaque to X-rays.

COMMON ION EFFECT:

The decrease in solubility of a sparingly soluble salt by addition of highly soluble salt with one ion common.

